



MAJICAST



DEVELOPERS

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Predicting contamination risk for water points across Kenya through machine learning, satellite data, and citizen science.

PROBLEM STATEMENT

💧 Background:

Clean water is a fundamental human right, yet over 2 billion people worldwide still lack access to safely managed drinking water.

Project Goal:

Develop a machine learning model that predicts contamination risk in rural water points using environmental and infrastructure data, reducing reliance on costly and time-consuming laboratory testing.

🌍 Impact:

Enable NGOs and governments to take proactive action, preventing public health crises and improving access to safe water.





OBJECTIVES

WHAT WE AIM TO ACHIEVE

- 
- Collect and preprocess geospatial, weather, and textual data for Kenyan rural communities.
- 
- Develop a dashboard or app for actionable insights
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- Build and validate a contamination risk prediction model.
- 
- Enable actionable intervention planning.
- 
- Integrate NLP for early-warning signals.

DATA SOURCES

- GEMS/UNEP Water Quality Monitoring (GEMStat)
- WPDx Kenya– Water Point Metadata
- Remote sensing – GEE (Google Earth Engine) – vegetation indices (population, temperature, ndvi, chirps)
- World Bank WASH Infrastructure Data
- NGO field reports (simulated or public reports)



MODELING OVERVIEW

WPDx attributes merged with environmental variables from Google Earth Engine. (WPDx+GEE)

➤ Structured Data Models:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM

➤ NLP Models:

- TF-IDF + SVM
- Fine-tuned DistilBERT

➤ Interpretability: SHAP and LIME for feature impact analysis





XGBOOST MODEL

- Risk score based on population, water pressure, and surface temp
- Risk levels: 0 (Low) to 3 (High)
- Balanced data using smart sampling
- Final model: XGBoost—high accuracy





GEMS DATASET & ISOLATION FOREST

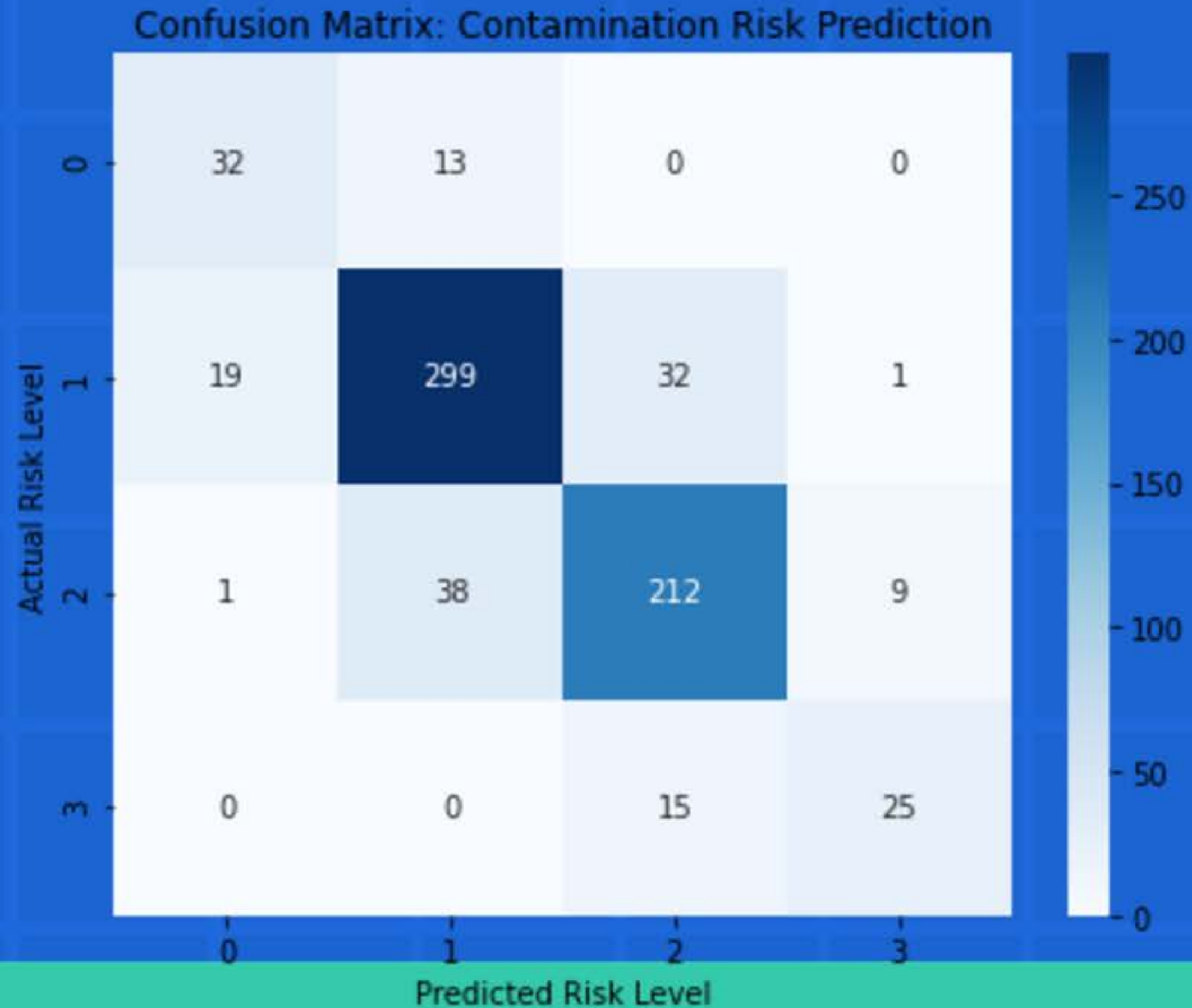
- GEMS – enhancing our data quality.
- Maintained the XGBoost model while adding GEMS to include chemical accuracy
- Trained an Isolation Forest: pH, temperature, conductivity.
- Enabled real-time anomaly detection even without expensive lab testing.





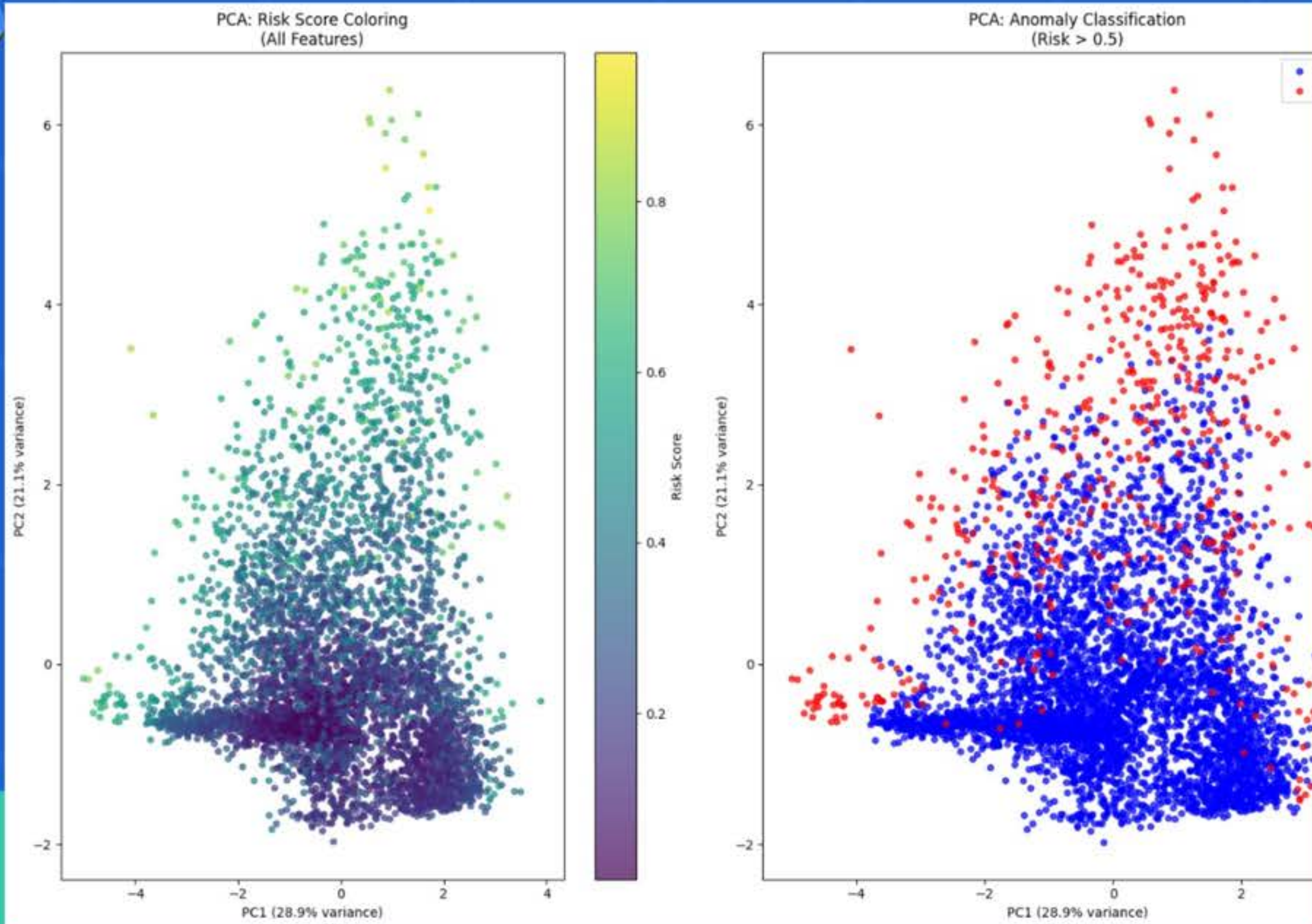
MODEL PERFORMANCE XGBOOST

- 81% overall accuracy
- 63% accuracy in predicting high-risk areas
- Supports targeted interventions



MODEL PERFORMANCE

ISOLATION FOREST



NLP MODEL FROM FIELD REPORTS

Text Preprocessing

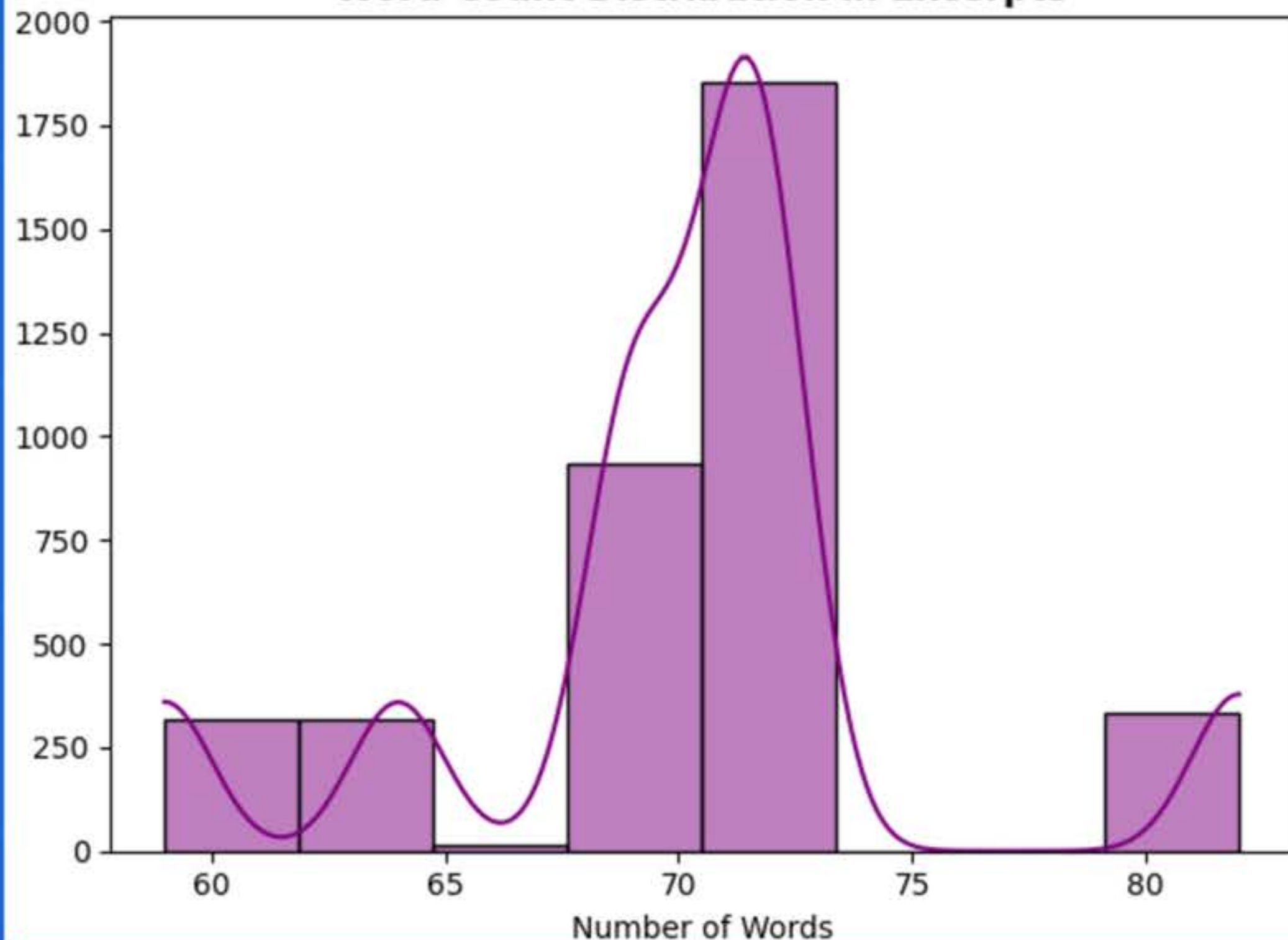
- Cleaned, tokenized, lemmatized
- Removed noise (stop words, digits, punctuation)

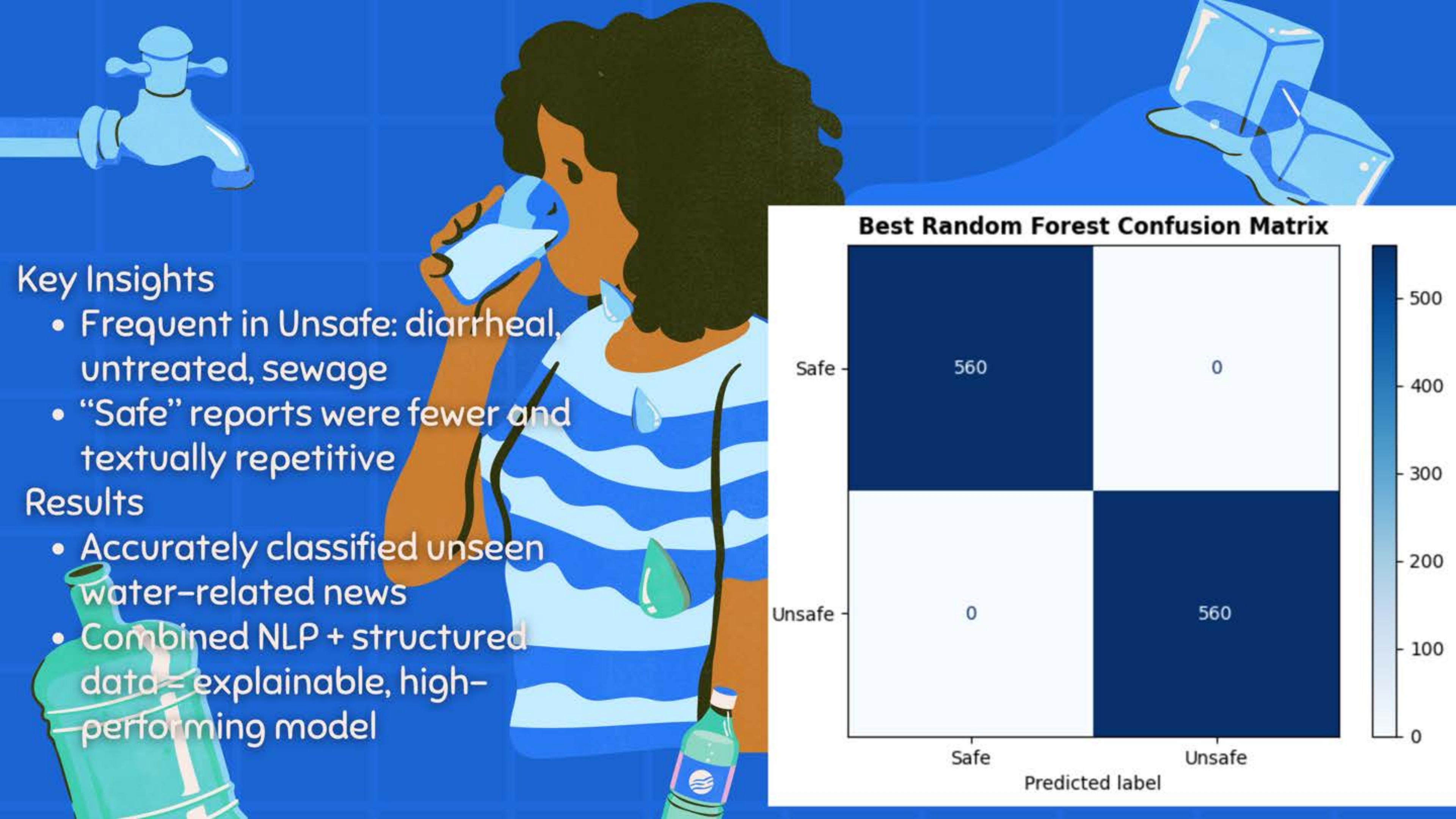
Feature Extraction

- TF-IDF vectorization on critical terms: contaminated, sewage, outbreak
- Binary classification: Safe vs. Unsafe



Word Count Distribution in Excerpts



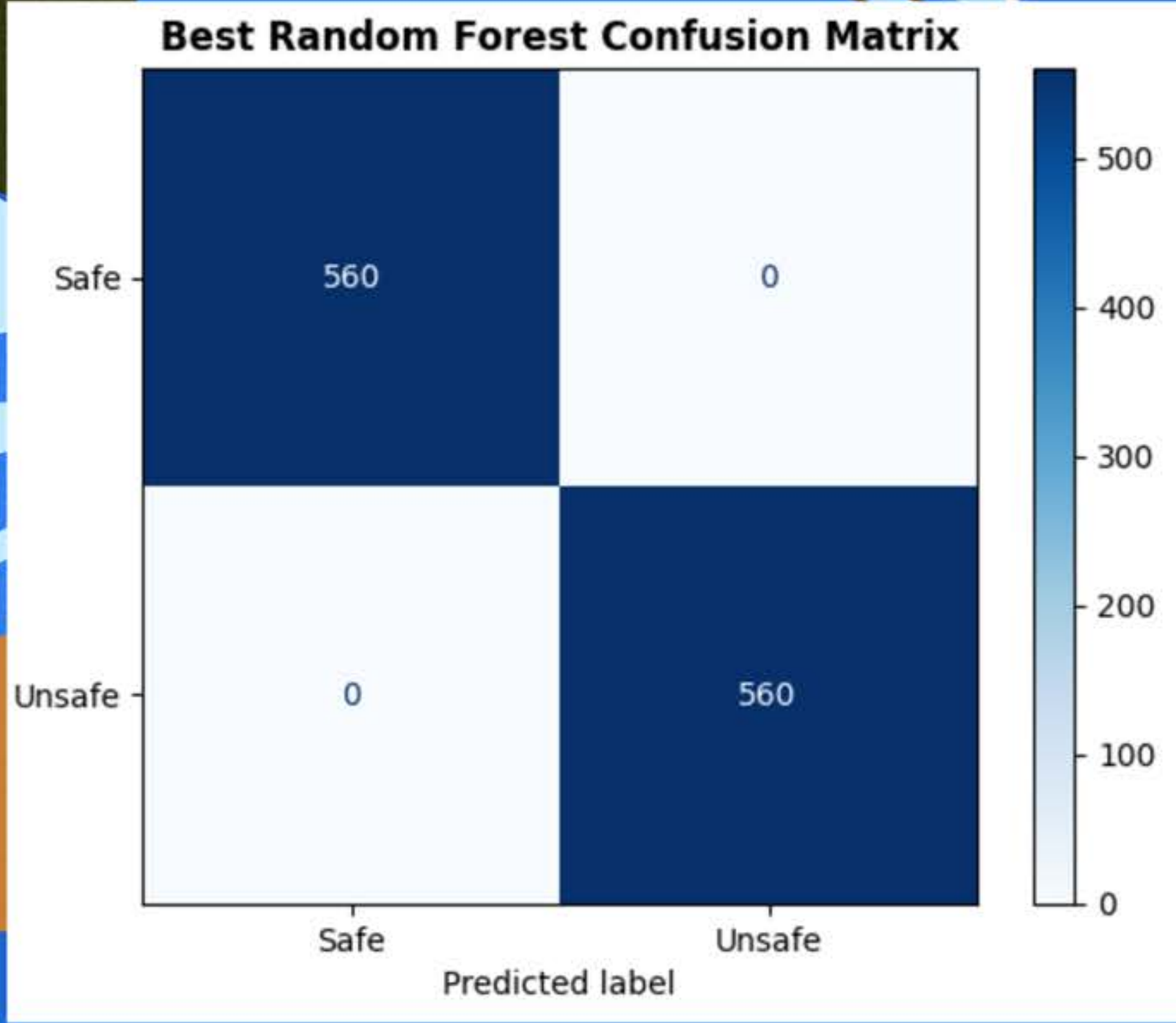


Key Insights

- Frequent in Unsafe: diarrheal, untreated, sewage
- “Safe” reports were fewer and textually repetitive

Results

- Accurately classified unseen water-related news
- Combined NLP + structured data = explainable, high-performing model



DEPLOYMENT

- Hosting: Frontend deployed via Vercel; Backend APIs deployed via Render.
- Frontend: Developed using Next.js (React 18) and TailwindCSS, offering a highly responsive, modern dashboard interface for water point risk tracking.
- Backend: Service-oriented architecture using a separate FastAPI server that loads XGBoost, IsolationForest, and NLP models via joblib/pickle to serve predictions on demand.

Future iterations will focus on adding persistent storage (PostgreSQL) and expanding the real-time sensor API capacity.



NEXT STEPS



- Integrate real-time geospatial feeds
- Expand model to other East African regions



- Collaborate with local water boards for on-ground validation
- Integrate live water crisis feeds using web scraping or APIs
- Expand Next.js app with multi-language support



- Fine-tune NLP with more field reports
- Incorporate user feedback into app



Challenges

- Data Quality Gaps: Incomplete metadata, uneven field reports
- Scalability: Backend and API constraints
- Model Bias: Underrepresented regions
- Limited Labels: Few verified contamination ground truths





Recommendations

- Collaborate with local NGOs for data collection
- Combine mobile + satellite data for more context
- Involve community feedback loops
- Prioritize mobile-first deployment



**EVERY DROP
COUNTS!**

THANK YOU

